# Vaccines



How Cowpox Scabs, Weakened Viruses,
Sugar Coats, and Fat Droplets
Taught the Body to Fight Back

MEDICINE & IMMUNOLOGY

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# Vaccines

*Teaching the Immune System With Harmless Mugshots Before the Real Fight*

Catching a dangerous infection for the first time is a race against the clock. A virus or bacterium can double its numbers inside your body every hour. Meanwhile, your immune system takes a full week to find the right defensive cells, multiply them, and build custom antibodies that fit the invader's outer shell. If the germ destroys your lungs, liver, or nerves before that first week is up, you lose the race.

Once you survive an infection, however, a team of veteran memory cells stays behind in your lymph nodes and bone marrow for decades. The next time that same germ enters your body, those memory cells recognize its outer coat within hours. They flood your blood with antibodies before you even feel a fever.

Every vaccine is a way to give your immune system that first practice lesson without putting you through the deadly disease. Over the past two centuries, vaccine makers have invented twelve clever disguises to show your body the enemy's mugshot safely. They have borrowed mild animal cousins, dried viruses in glass flasks, and locked bacterial poisons shut with formaldehyde. They have walked jungle viruses through hundreds of eggs, brewed empty protein shells in baker's yeast, tied slippery bacterial sugars to proteins so babies can spot them, and pinned shape-shifting viral spikes inside microscopic fat bubbles.

CHAPTER 1

# Smallpox and the Cowpox Cousin

*How Milkmaid Blisters, Orphan Chains, and Split Needles Wiped Out a Human Plague*

Across rural England in the 1700s, country dairymaids were famous for having clear, unscarred skin. At the time, smallpox killed one in three infected children and left survivors blinded or deeply pitted for life. Milkmaids regularly caught cowpox, a mild infection from blisters on cows' udders that left a few sores on their hands for a week. Once those sores healed, the milkmaids almost never caught smallpox.

In May 1796 in Berkeley, Gloucestershire, country doctor Edward Jenner put this farmyard observation to a test. He took a drop of fluid from a cowpox blister on the hand of milkmaid Sarah Nelmes. He scratched that fluid into the arm of a local boy named James Phipps. Phipps ran a mild fever for a single day and recovered. Six weeks later, Jenner scratched live smallpox pus into the boy's arm, and not a single blister formed.

Why did a mild cow virus protect a human against a killer plague? Smallpox and cowpox belong to the same family of poxviruses. Because they evolved from a shared ancestor, both viruses wear nearly identical protein coats on their outer surfaces. When cowpox multiplied harmlessly in the skin of Phipps's arm, his immune system built antibodies and memory cells shaped to grab those surface proteins. When real smallpox arrived six weeks later, it wore the same outer uniform. Phipps's trained antibodies swarmed and destroyed the virus before it could spread.

Getting that live virus across oceans before refrigerators existed required an extraordinary human relay. Dried cowpox threads often died on long sea voyages in tropical heat. In November 1803, Spanish physician Francisco Javier de Balmis sailed from A Coruña aboard the ship *María Pita* with twenty-two orphan boys between ages three and nine. None of the boys had ever caught cowpox or smallpox. At departure, Balmis scratched cowpox into the arms of two boys. Nine days later, just as their arm blisters ripened, he transferred fresh fluid into the next pair of boys. That living chain of blisters crossed the Atlantic and Pacific, bringing vaccination to hundreds of thousands of people across the Americas and Asia.

Yet smallpox still killed two million people a year in the 1960s until two simple tools finished the job. First, in 1965 at Wyeth Laboratories, microbiologist Benjamin Rubin ground down the eyelet of a sewing machine needle into a tiny two-pronged fork. Dipped into a vaccine vial, the slot between the two prongs held just 0.0025 milliliters of liquid by surface tension. That tiny drop was one-fourth the old syringe dose, stretching global vaccine supplies fourfold. Any village volunteer could learn in five minutes to dip the fork and jab a patient’s arm fifteen times.

Second, epidemiologists William Foege and Donald Henderson realized they did not need to vaccinate every person on Earth at once. Instead, they used the mathematics of the herd immunity threshold $p_c$:

$$p_c = 1 - \frac{1}{R_0}$$

In an unvaccinated town, each smallpox patient infected roughly $R_0 = 5$ other people. Once $p_c = 1 - 1/5 = 80\%$ of the people around

an outbreak were immune, each sick person infected fewer than one new person, and the chain of transmission died out. World Health Organization teams tracked down every new smallpox patient, isolated them at home, and vaccinated a tight ring of family, neighbors, and coworkers around each village. Because smallpox could not hide in wild animals, that ring strategy cornered the last natural case on Earth—hospital cook Ali Maow Maalin in Merca, Somalia—in October 1977.

CHAPTER 2

# Rabies and the Dried Spinal Cord

*How Louis Pasteur Raced a Lethal Virus on Its Slow Climb Up the Nerves*

Once a virus has already entered your body, giving a vaccine is almost always too late. Training new antibodies from scratch takes ten to fourteen days, and most viruses multiply through your bloodstream in hours. Rabies is the famous exception to this rule because of the strange, slow way the virus travels. Instead of jumping into your blood after a dog bite, rabies virus enters the tips of motor nerves in the torn muscle. It then crawls upward along nerve fibers toward your spinal cord and brain at a snail's pace of just one to two millimeters per hour.

If a rabid dog bites your hand or leg, the virus can take anywhere from three weeks to three months to reach your brain. Once it arrives in the brain and symptoms start, rabies is nearly 100 percent fatal. While the virus is still creeping up the nerves in your arm or leg, however, it hides inside nerve fibers without waking up your immune system. Louis Pasteur realized that if he injected a vaccine right after a bite, the immune system could build a wall of antibodies in ten days and ambush the virus before it reached the spinal cord.

In the 1880s, microscopes could not see viruses, and nobody knew how to grow them in laboratory dishes. Pasteur and his colleague Émile Roux had to grow rabies inside the nervous systems of rabbits. Passing the virus from rabbit to rabbit produced a predictable strain that always reached the rabbit's brain in seven days. To weaken that lethal virus into a safe training dose, Roux and Pasteur removed the spinal cords of infected rabbits. They hung

the cords by threads inside sterile glass flasks with dry potassium hydroxide crystals at the bottom:

- **Fourteen days of drying**: The crystals soaked up moisture while oxygen in the flask slowly damaged the virus. After fourteen days of drying, a rabbit spinal cord lost all ability to cause rabies, leaving behind dead viral proteins that safely showed the immune system the virus's outer shape.
- **Stepping up the ladder**: Each day, a patient received an injection of mashed spinal cord that had dried one day less—twelve days, ten days, seven days—gradually waking up stronger immune defenses as antibodies rose.
- **The live challenge test**: By day ten, the patient could survive an injection of fresh, one-day-dried spinal cord that would kill an unprotected animal, proving their blood was packed with protective antibodies.

In July 1885, a young boy named Joseph Meister arrived in Paris from Alsace at Pasteur's laboratory on Rue d'Ulm. A rabid dog had knocked the boy down and bitten him fourteen times on his hands, legs, and thighs. Because Pasteur was a chemist and not a licensed medical doctor, he asked physicians Alfred Vulpian and Jacques Grancher to examine the boy's deep wounds. Together they agreed the boy was certain to die without treatment. Grancher gave Meister the first injection of fourteen-day-dried rabbit cord that evening. Over ten tense days, Meister received thirteen injections stepping all the way up to fresh rabid cord. His immune system won the race against the nerve-climbing virus. Meister never developed rabies, grew up healthy, and worked for decades as the gatekeeper of the Pasteur Institute in Paris. Today, four painless shots of killed cell-culture rabies virus paired with ready-made antibodies injected straight into the bite wound save nearly 100 percent of patients treated promptly.

CHAPTER 3

# Diphtheria, Tetanus, and the Disarmed Poison

*How Gaston Ramon Used Formaldehyde and Kitchen Tapioca to Tame Two Bacterial Toxins*

Most germs make you sick by multiplying across your organs, but the bacteria behind diphtheria and tetanus barely travel at all. Diphtheria bacteria camp inside a leathery gray patch at the back of a child's throat. Tetanus bacteria hide inside a deep, airless puncture wound from a rusty nail or garden thorn. From those tiny hideouts, both bacteria pump out a protein poison—a toxin—that enters the bloodstream. Diphtheria toxin shuts down protein factories inside heart and nerve cells. Tetanus toxin blocks the nerve switches that let muscles relax. Within days, it locks the jaw and chest in fatal spasms. Just one-millionth of a gram of tetanus toxin is enough to kill an adult human.

In 1890 in Berlin, Emil von Behring and Shibasaburo Kitasato discovered that injecting animals with tiny drops of these bacterial broths caused their blood to fill with antitoxin antibodies. Giving horse antitoxin serum to a child choking with diphtheria saved thousands of lives. Yet there was a big catch. Borrowed horse antibodies were cleared from a child's blood in three weeks, leaving no lasting memory cells behind. Worse, a second injection of horse blood often triggered severe allergic shock. For permanent protection, human immune systems needed to practice on the poison itself without dying from it.

In 1923 at the Pasteur Institute annex in Garches outside Paris, veterinarian Gaston Ramon noticed something useful. A batch

of diphtheria toxin treated with formalin—water mixed with formaldehyde to keep stray germs out—had lost its ability to kill guinea pigs. Ramon found that warming diphtheria or tetanus toxin with 0.3 to 0.4 percent formaldehyde at body temperature (37 degrees Celsius) for four weeks stitched tiny chemical bridges across the poison protein. Those bridges bolted the poison's cutting blade shut while leaving its outside bumps unchanged. Ramon called this disarmed poison a "toxoid." When injected into a child, the harmless toxoid trained antibodies that grabbed the real poison the instant it appeared.




FIG. 3.1 — Formaldehyde cross-links lock the bacterial poison shut while aluminum salts ring the immune alarm.

While harvesting antitoxin from horses at Garches in 1925, Ramon noticed a second puzzle. Horses that developed a small, sterile lump of inflammation at the needle site made far more antibodies than horses whose skin stayed calm. A pure, clean protein often slips past the immune system unnoticed because it causes no tissue damage and rings no alarm bells. Ramon tested mixing his toxoid with breadcrumbs, olive oil, and ordinary kitchen tapioca starch from his pantry. Antibody levels jumped severalfold.

A year later in London, chemist Alexander Glenny achieved the same boost with a cleaner ingredient. He precipitated diphtheria

toxoid onto tiny crystals of aluminum potassium sulfate, or alum. Those microscopic aluminum salts held the toxoid at the injection site like a slow-release sponge and irritated local muscle cells just enough to call immune scout cells to the spot.

Combined with killed whooping cough bacteria in the 1940s as the childhood DTP shot, diphtheria and tetanus toxoids transformed child health. Even better, when a pregnant mother receives a tetanus shot, her antibodies cross the placenta into her baby's blood. That borrowed shield protects newborns when umbilical cords are cut in unsterile conditions, cutting global newborn tetanus deaths from roughly 787,000 babies a year in 1988 to fewer than 25,000 a year within three decades.

CHAPTER 4

# Yellow Fever and the Hundredth Egg

*How Max Theiler Walked a Jungle Virus Through Mouse Brains and Chick Embryos*

In a Rockefeller Foundation laboratory freezer in New York in 1936 sat a glass vial labeled 17D. Inside floated a strain of yellow fever virus descended from a West African outbreak nine years earlier. It had been passed from dish to dish so many times that it had forgotten how to attack a human liver or ride inside a mosquito.

Yellow fever is a mosquito-borne virus that destroys liver cells, turning the skin yellow with jaundice, causing dark internal bleeding, and killing up to half of severe patients. For centuries, outbreaks emptied port cities from Havana and New Orleans to Lagos and Rio de Janeiro. In June 1927, Rockefeller researchers Adrian Stokes, Johannes Bauer, and N. Paul Hudson working in Lagos, Nigeria received blood drawn from a Ghanaian patient named Asibi who survived a mild fever. They injected Asibi's blood into an Indian rhesus monkey and isolated the yellow fever virus for the first time. The work was perilous. Two months later, Adrian Stokes died in Lagos after catching yellow fever in the lab, one of six Rockefeller scientists killed by the virus they were trying to stop.

Working with monkeys was slow, expensive, and deadly. In 1930, South African physician Max Theiler—working first at Harvard and then at the Rockefeller Institute in New York—found a simpler animal host. He discovered that the Asibi virus multiplied readily when injected into the brains of white laboratory mice. Passing the virus from mouse brain to mouse brain weakened its ability to damage monkey livers. Unfortunately, living in mouse

brains taught the virus to attack nerve tissue instead. That side effect made the mouse-adapted virus too dangerous to inject into humans without risking fatal brain inflammation.

To strip out both liver damage and nerve damage at the same time, Theiler and colleague Hugh Smith forced the virus to evolve inside tissue that had neither liver nor brain cells:

- **Removing the nerve target**: In flask series 17D, Theiler and Smith grew the virus in minced chicken embryos from which they had carefully snipped out the brain and spinal cord with scissors before mincing.
- **Evolution by trade-off**: Because a virus carries only a tiny genetic playbook, mutations that helped it multiply rapidly inside headless chick embryo cells over more than a hundred transfers broke the molecular keys it needed to invade human liver cells and nerve cells.
- **Mosquito lock-out**: Between the 89th and 114th flask transfers in 1936, the 17D strain lost its toxicity completely. In human volunteers, a tiny under-the-skin dose multiplied just enough to build lifelong antibodies, and it could no longer multiply inside *Aedes aegypti* mosquitoes, ensuring mosquitoes could never spread the vaccine virus back into the wild.

How lucky was that exact evolutionary path inside flask 17D? When Theiler and Smith tried to repeat the experiment from scratch with fresh Asibi virus, the virus never mutated the same way again. Modern genetic sequencing later revealed that thirty-two small mutations across the viral genome separate 17D from the original Asibi strain. Every dose of yellow fever vaccine given over the past eighty-five years—more than eight hundred million doses brewed inside fertilized chicken eggs—is a direct descendant of that single 1936 17D flask, earning Max Theiler the 1951 Nobel Prize.

CHAPTER 5

# Polio and the Killed-Versus-Live Rivalry

*How Jonas Salk's Formaldehyde Bath and Albert Sabin's Sugar Cube Stopped Summer Paralysis*

Every summer in the mid-twentieth century, swimming pools, beaches, and movie theaters across North America closed as parents dreaded poliomyelitis. Polio is a hardy virus that enters through the mouth and multiplies inside the intestines. In roughly 99 percent of children, it causes only a mild summer fever or no symptoms at all. In about one out of two hundred infections, however, the virus crosses into the spinal cord and destroys the motor nerves that move the legs and chest, leaving children in leg braces or iron-lung breathing tanks.

Early attempts to make a polio vaccine in 1935 using ground-up monkey spinal cords ended in tragedy, paralyzing several children. The breakthrough came in 1949 at Boston Children's Hospital when John Enders, Thomas Weller, and Frederick Robbins proved that poliovirus did not need nerve cells to grow. It multiplied cleanly in glass flasks of skin, muscle, and kidney tissue. Once vats of poliovirus could be grown in monkey kidney cells without dangerous nerve tissue, two fierce rivals took opposite paths to a vaccine.

At the University of Pittsburgh, Jonas Salk believed you did not need to suffer a live infection to gain immunity. Salk grew all three types of poliovirus, filtered the liquid to remove cell clumps, and soaked the virus in a 1-to-4,000 bath of formalin at 37 degrees Celsius. Salk tracked how the concentration of live virus $N(t)$ dropped over time $t$ as formaldehyde cross-linked its outer shell along a first-order exponential curve:

$$N(t) = N_0 e^{-kt}$$

Even after the concentration of detectable live virus hit zero around day six, Salk kept cooking the batch in formaldehyde for another six days. That extra margin drove the mathematical chance of a single surviving virus particle far below one in a million doses while keeping the outer shell intact. Announced by Thomas Francis Jr. on April 12, 1955 after a field trial in 1.8 million schoolchildren, Salk's injected killed shot filled the bloodstream with antibodies (IgG) that blocked poliovirus from ever reaching the spinal cord. (When Cutter Laboratories failed to filter out microscopic cell clumps that same month, live virus hiding inside the clumps survived the formaldehyde bath and paralyzed two hundred children before strict filtration rules fixed the process permanently.)

At the University of Cincinnati, Albert Sabin argued that an injected killed vaccine left a loophole. It protected a child's spinal cord by putting antibodies in the blood, but it did not stop wild poliovirus from multiplying in the child's intestines and passing to neighbors through sewage. Sabin bred weakened live strains of all three polio types that could be swallowed in a drop of pink liquid on a sugar cube. Because Sabin's weakened virus multiplied harmlessly inside the lining of the gut, it trained local antibodies (secretory IgA) right on the intestinal wall. That gut shield slammed the front door shut so vaccinated children could neither get paralyzed nor spread polio to anyone else.

Yet Sabin's live virus carried a tiny risk of mutating back toward paralysis in roughly one out of 2.4 million doses. Since the Global Polio Eradication Initiative began in 1988, combining Sabin's oral drops to stop community outbreaks with Salk's injected shot to lock in individual safety has cut polio paralysis from 350,000 chil-

dren a year across 125 countries to fewer than a hundred cases in two countries, with wild types 2 and 3 eradicated worldwide.

CHAPTER 6

# Measles, Mumps, and the Memory Eraser

*How Maurice Hilleman Brewed a Chick-Embryo Trio That Protects Your Old Immune Memories*

For decades, doctors tracking childhood survival noticed a puzzling bonus whenever a measles vaccine campaign arrived in a region. When measles shots rolled out across England, the United States, and West Africa, deaths from measles dropped to near zero as expected. At the same time, however, childhood deaths from completely unrelated infections—like bacterial pneumonia, ear infections, and severe diarrhea—also fell by up to 50 percent. Why would a measles shot protect a child against bacteria and other viruses that look nothing like measles?

Before vaccines, measles infected nearly every child before age fifteen. It spreads through the air in tiny droplets that hang in a room for two hours after a sick person leaves, allowing one infected person to spread measles to twelve to eighteen unvaccinated people. In February 1954, pediatricians Thomas Peebles and John Enders isolated the virus in Boston from throat swabs and blood of a schoolboy named David Edmonston during an outbreak at the Fay School. Enders weakened the virus in chick cells to create the first measles vaccine in 1963, though it still gave many children a high fever and rash.

At Merck Laboratories in West Point, Pennsylvania, microbiologist Maurice Hilleman—who grew up raising chickens on a Montana farm—passed the Edmonston strain forty additional times through chick embryo cells kept at a cool 32 degrees Celsius. That extra cooling bred the much gentler Moraten (“More Attenuated Enders”) strain licensed in 1968. When his own young daughter

Jeryl Lynn woke him in March 1963 with a swollen jaw from mumps, Hilleman drove to his office at midnight to fetch throat swabs. He grew her virus in fertilized hen's eggs to create the Jeryl Lynn mumps vaccine within four years, and in 1971 he combined both with Stanley Plotkin's rubella strain into the single MMR shot:

- **Measles (Moraten strain)**: Grown in cool chick embryo cells so it barely causes fever while raising lifelong neutralizing antibodies in over 97 percent of two-dose recipients.
- **Mumps (Jeryl Lynn strain)**: Adapted from his daughter's throat swab in hen's eggs until it lost the ability to inflame the brain, salivary glands, or reproductive organs.
- **Rubella (RA 27/3 strain)**: Grown by Stanley Plotkin in human cells at 30 degrees Celsius to stop rubella from crossing the placenta in pregnant women and causing blindness, deafness, and heart defects in unborn babies.

The mystery of why measles vaccination saved children from unrelated diseases was finally solved between 2012 and 2019 by immunologists Rik de Swart, Michael Mina, and Stephen Elledge. Unlike ordinary cold viruses that infect nose cells, wild measles virus uses a surface hook (CD150) that sits specifically on your immune system's own memory B-cells and memory T-cells. Those are the veteran cells in your lymph nodes and bone marrow that remember every cold, flu, and bacterial infection you survived since infancy.

In 2019, Mina and Elledge sequenced the entire antibody library in the blood of seventy-seven unvaccinated Dutch children before and two months after a natural measles infection. They proved that wild measles physically hunts down and kills existing memory cells, wiping out between 11 and 73 percent of a child's pre-existing antibodies. For two to three years after surviving measles,

a child's immune memory is erased like a formatted hard drive, leaving them defenseless against germs they had already beaten until the MMR vaccine keeps the memory eraser out.

CHAPTER 7

# Hepatitis B and the Baker's Yeast Decoy

*From Baruch Blumberg's Blood Plasma Filter to the First Cancer-Preventing Vaccine Brewed in Vats*

In 1965 at the Fox Chase Cancer Center in Philadelphia, geneticist Baruch Blumberg discovered a mysterious protein in blood samples that turned out to be the outer surface coat of the hepatitis B virus. Spread through blood and from mother to baby at birth, hepatitis B silently scars the liver over decades and causes more than half of the world's primary liver cancers. When Blumberg and colleagues examined the blood of chronic hepatitis B carriers under an electron microscope in 1968, they spotted a strange quirk in how the virus builds itself.

Inside an infected liver cell, the complete infectious hepatitis B virus is a 42-nanometer sphere packed with viral DNA. Yet for every single live virus particle the liver cell releases, it also dumps a thousand tiny, 22-nanometer empty spheres into the bloodstream. Those miniature spheres are made solely of the virus's outer surface protein (HBsAg) with zero viral DNA inside. The virus pumps out those empty decoys as a smoke screen to soak up the patient's antibodies while the real 42-nanometer viruses slip past and infect fresh liver cells.

Because hepatitis B virus refused to grow in laboratory cell dishes, Blumberg and colleague Irving Millman realized in 1969 that the virus's own smoke screen could be turned against it. If they filtered those harmless 22-nanometer empty spheres out of the blood of chronic human carriers, they could inject the decoys into healthy people to train their immune systems in advance. At Merck, Maurice Hilleman collected blood plasma

from chronic carriers and spun it in high-speed centrifuges to separate the tiny empty spheres from the larger live viruses. To be certain no live virus or stray pathogen survived, Hilleman's team treated the spheres with three harsh chemical steps—digestive pepsin enzyme at pH 2, urea to break apart stray proteins, and formaldehyde—producing the first plasma hepatitis B vaccine in 1981.

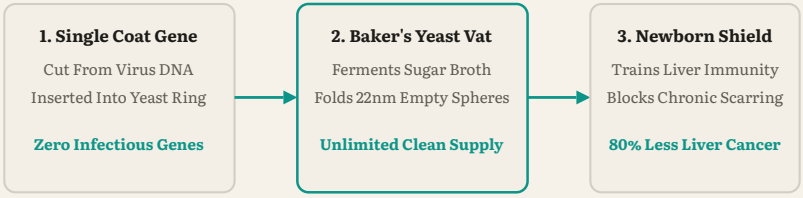



FIG. 7.1 — Baker's yeast vats brew harmless 22-nanometer empty protein shells without a drop of human blood.

Purifying vaccine from human blood plasma worked, but it had major drawbacks. Each batch took sixty-five weeks to manufacture and safety-test in chimpanzees. It also relied on a limited pool of human blood donors and made doctors and patients nervous as the HIV epidemic emerged in the early 1980s. In 1981, biochemist William Rutter at the University of California, San Francisco, Pablo Valenzuela at Chiron Corporation, and geneticist Benjamin Hall at the University of Washington found a way to retire human blood forever.

They snipped out the single gene that codes for the hepatitis B surface protein and stitched it into a circular DNA plasmid inside ordinary baker's yeast (*Saccharomyces cerevisiae*)—the same single-celled fungus used to bake bread and brew beer. When

the yeast multiplied inside stainless-steel fermentation tanks fed with sugar and nutrients, the yeast cells built pure viral surface proteins that automatically curled up into the exact same 22-nanometer empty spheres. Because only one coat gene was copied into the yeast, it was physically impossible for the vat to produce a live virus. Approved by the Food and Drug Administration in July 1986 as the world's first genetically engineered human vaccine, it also proved that a vaccine could prevent human cancer: after Taiwan launched universal newborn hepatitis B vaccination in July 1984, chronic liver infection in children plunged from 10 percent to under 0.6 percent, and childhood liver cancer rates dropped by 75 to 80 percent.

CHAPTER 8

# Hib and the Sugar-Coat Trick

*How John Robbins and Rachel Schneerson Tied a Slippery Bacterial Coat to a Protein So Babies Could See It*

Until the late 1980s, pediatric emergency rooms across the world dreaded a bacterium called *Haemophilus influenzae* type b, or Hib. Despite its name—given during the 1890 flu pandemic before scientists knew flu was a virus—Hib is a fast-moving bacterium that lives in the nose and throat. Before vaccines, it invaded the blood and spinal fluid of roughly 20,000 American children under age five every year. It was the leading cause of bacterial meningitis in infants, killing one in twenty infected babies, leaving a quarter of survivors with permanent deafness or brain damage, and swelling the throat flap so rapidly that toddlers could suffocate within hours.

In the 1970s, researchers purified the outer coat of the Hib bacterium and tested it as a vaccine in a massive trial of 100,000 children in Finland. A baffling age split appeared in the trial results. The vaccine protected children over two years old and adults, but in babies under eighteen months—the exact age group suffering 80 percent of serious Hib disease—it failed to protect them at all.

Why were babies' immune systems blind to the vaccine? Unlike viruses, which wear protein coats, dangerous bacteria like Hib, pneumococcus, and meningococcus hide inside a slick outer capsule made of long sugar chains called polysaccharides. Your immune system has two branches that must work together for lasting protection. B-cells make antibodies, while helper T-cells act as coaches that tell B-cells to multiply, upgrade their antibodies from weak early versions to high-strength versions, and form

long-term memory. Here is the catch: helper T-cells can only see pieces of *protein* they are completely blind to pure *sugars*. An adult has specialized B-cells in the spleen that can make modest, short-lived antibodies against sugars on their own, but a baby's spleen does not finish building that zone until around age two.

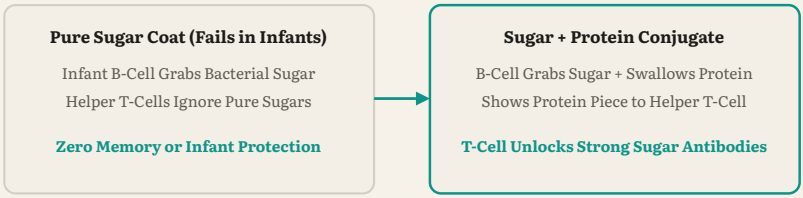



FIG. 8.1 — Tying the bacterial sugar coat to a harmless protein tricks helper T-cells into coaching infant B-cells.

In 1980 at the National Institutes of Health in Bethesda, Maryland, physicians John Robbins and Rachel Schneerson—alongside Porter Anderson and David Smith at the University of Rochester—revived a forgotten 1929 rabbit experiment by Oswald Avery and Walther Goebel. They turned it into a molecular bait-and-switch called a "conjugate" vaccine. They took the slippery sugar chain from the outside of the Hib bacterium and chemically tied it with a covalent bond to a harmless carrier protein that T-cells love to recognize, such as tetanus or diphtheria toxoid.

When a young baby's B-cell spots that linked molecule, its surface sensor grabs the Hib sugar coat and swallows the whole package—sugar and attached protein together. Inside the B-cell, enzymes chop up the attached carrier protein and display those protein pieces on the cell's outer window. A passing helper T-cell recognizes that protein fragment, latches onto the B-cell, and coaches

it to pump out high-strength antibodies aimed straight at the Hib sugar coat.

Once conjugate Hib vaccines were approved for infants starting at two months of age in December 1990—and the same sugar-to-protein trick was later applied to pneumococcal and meningococcal bacteria—invasive Hib meningitis in young children plunged by more than 99 percent within five years. Because vaccinated babies also stopped carrying the bacteria in their noses, the shots created herd protection for newborns too young for their first dose, turning a weekly emergency-room tragedy into a medical rarity.

CHAPTER 9

# HPV and the Hollow Virus Shell

*How Jian Zhou and Ian Frazer Built Self-Assembling Empty Shells That Stop Cervical Cancer*

In 1983 and 1984 at the German Cancer Research Center in Heidelberg, virologist Harald zur Hausen overturned decades of medical belief by proving that cervical cancer—one of the leading cancer killers of women worldwide—is caused by a persistent viral infection. Using DNA probes on cervical tumor biopsies, zur Hausen isolated human papillomavirus types 16 and 18 (`HPV-16` and `HPV-18`), which account for roughly 70 percent of all cervical cancers. He showed that when these viruses linger inside cervical cells for years, two of their genes (`E6` and `E7`) disable the cell's natural tumor-suppressor brakes (`p53` and `Rb`) and drive cancerous growth.

Knowing a virus caused the cancer immediately raised the hope of a preventive vaccine. Yet vaccine makers hit two walls at once. First, HPV refused to grow inside ordinary laboratory cell cultures because the virus only builds its outer shell inside maturing layers of living human skin or mucosal tissue. Second, nobody dared inject a killed or weakened whole HPV virus into healthy children and teenagers when the virus's DNA carried two active cancer-causing genes.

At the University of Queensland in Brisbane, Australia in 1991, virologist Jian Zhou, molecular biologist Xiao-Yi Sun, and immunologist Ian Frazer—followed independently by Douglas Lowy, John Schiller, and Richard Schlegel at the U.S. National Cancer Institute and Georgetown University—discovered a shortcut built into the physics of the virus's outer shell. The outside of a papillomavirus is a microscopic soccer ball built from 72 five-sided

stars made of a single structural protein called L1. When Zhou and Sun inserted the gene for L1 alone into cells in late 1991 and looked through an electron microscope, they saw that 360 copies of the L1 protein had automatically snapped together into hollow 55-nanometer spheres. Those spheres looked identical to live HPV from the outside, while remaining completely empty on the inside:

- **Self-assembling geometry**: Just as soap bubbles lock into spheres automatically, L1 proteins brewed inside baker's yeast or insect cells fold into five-sided blocks that click into a 72-block hollow shell without needing a single strand of viral DNA.
- **The spacing alarm**: Loose single proteins barely wake up B-cells. By contrast, a hollow Virus-Like Particle (VLP) displays hundreds of identical bumps spaced five to ten nanometers apart—the exact spacing that grabs dozens of sensors on a B-cell at once and rings a five-alarm fire bell in the lymph nodes.
- **Forty-fold antibody flood**: Natural HPV stays hidden inside the outer skin layer where blood vessels are scarce, so it barely alerts your immune system. Because the injected hollow shells enter the arm muscle and lymph nodes directly, vaccinated teenagers build blood antibody levels up to forty times higher than people who recover from a natural HPV infection.

Those surging blood antibodies seep into the fragile lining of the cervix and coat any arriving HPV virus the moment it tries to enter a basal cell. Licensed in 2006 (Gardasil and Cervarix) and later expanded to cover nine cancer-causing and wart-causing HPV types, these hollow-shell vaccines delivered results that exceeded even their inventors' hopes. National cancer registry studies in Sweden, England, and Scotland tracking millions of women through 2020 and 2024 showed that girls vaccinated before age fourteen experienced an 87 to 90 percent drop in invasive cervical cancer—

and in the Scottish cohort of women immunized at ages twelve to thirteen, researchers recorded zero cases of cervical cancer.

CHAPTER 10

# Rotavirus and the Cow-Human Swap

*How Paul Offit, Fred Clark, and Stanley Plotkin Shuffled Eleven RNA Strands to Save Infants From Dehydration*

In 1973 at the Royal Children's Hospital in Melbourne, Australia, microbiologist Ruth Bishop and her colleagues examined tiny tissue samples from the small intestines of babies hospitalized with severe winter diarrhea. Under the electron microscope, they saw clusters of 70-nanometer particles shaped like wagon wheels with short spokes and a smooth rim. They named the germ *rotavirus*, from the Latin *rota* for wheel. Nearly every child on Earth caught rotavirus before age five. In countries without ready access to intravenous fluids, the resulting dehydration killed more than 500,000 babies every year.

Because rotavirus attacks the cells lining the small intestine, an effective vaccine had to be swallowed by mouth so it could train protective antibodies right inside a baby's gut. In 1998, the first oral rotavirus vaccine (`RotaShield`), built from a monkey rotavirus strain, won approval. Just ten months later in 1999, it was pulled off the market after safety trackers linked it to intussusception—a dangerous telescoping fold of the infant bowel—in roughly 1 out of every 10,000 vaccinated babies.

Fortunately, starting in 1980 at the Children's Hospital of Philadelphia and the Wistar Institute, virologists Fred Clark, Paul Offit, and Stanley Plotkin had been building a much gentler vaccine using a genetic card trick called "reassortment." Unlike most viruses that store their genetic code on one continuous ribbon, a rotavirus splits its genome into eleven separate segments of double-stranded

RNA. Those eleven strands sit inside the viral shell like eleven playing cards in a deck, each coding for a different viral protein.

Clark, Offit, and Plotkin isolated a calf rotavirus strain (WC3) from a calf on a dairy farm in Chester County, Pennsylvania. Because the calf virus was adapted to cattle, it multiplied poorly in human babies and never made them sick. However, its outer coat looked too different from human rotavirus to give babies strong protection on its own.

To get cow safety and human protection in the same particle, the researchers infected a dish of monkey kidney cells with both the harmless calf rotavirus and a human rotavirus at the same time. Inside the shared cell, the two viruses shuffled their eleven RNA cards together, capable of producing $2^{11} = 2,048$ possible hybrid combinations. By screening the offspring with antibodies, the Philadelphia team picked out five exact hybrid viruses (RotaTeq). Each hybrid carried ten internal genes from the harmless cow virus for safety and a single outer-coat gene (G1, G2, G3, G4, or P1A[8]) from a human rotavirus so the baby's gut would learn to block the five major human strains.

To prove that the cow-human hybrids did not cause the bowel-folding side effect that doomed the monkey vaccine, Merck ran one of the largest infant trials in history (REST, 2001–2005). They enrolled 70,301 babies across eleven countries and measured vaccine efficacy $VE$ by comparing the attack rate in vaccinated infants ($AR_v$) against unvaccinated infants ($AR_u$):

$$VE = 1 - \frac{AR_v}{AR_u}$$

Across those 70,301 infants, the three-dose oral liquid showed zero increase in bowel folding. It cut severe rotavirus illness by

98 percent and reduced rotavirus hospitalizations and emergency-room visits by $VE = 94.5\%$. Licensed in February 2006 alongside a companion oral vaccine (`Rotarix`), rotavirus drops have since prevented hundreds of thousands of infant deaths and millions of hospital admissions worldwide.

CHAPTER 11

# Ebola and the Trojan Horse Ring

*How Heinz Feldmann Dressed a Harmless Livestock Virus in Ebola's Coat to Stop an Outbreak in Guinea*

For nearly forty years after Ebola virus was first identified along the Ebola River in 1976, outbreaks followed a grim script. The virus spilled over from forest wildlife, spread through families and hospital wards via direct contact with body fluids, and killed between 50 and 90 percent of infected patients before burning out. Because outbreaks were small, remote, and unpredictable, commercial drug companies saw little financial incentive to fund expensive human trials. As a result, no vaccine had completed clinical testing when Ebola erupted across Guinea, Liberia, and Sierra Leone in late 2013 and 2014, infecting more than 28,000 people and killing over 11,300.

A decade earlier, between 2001 and 2005 at Canada's National Microbiology Laboratory in Winnipeg, virologists Heinz Feldmann, Ute Ströher, and Steven Jones—using a viral rescue system created by John Rose at Yale—had built a Trojan-horse vaccine inside a high-containment lab. Working with live Ebola virus was far too dangerous for a weakened vaccine, and killed Ebola virus failed to protect monkeys. Instead, the Winnipeg team started with Vesicular Stomatitis Virus (VSV), a bullet-shaped livestock virus that causes mild mouth blisters in cattle and horses but produces at most a brief, harmless fever in humans.

Using genetic scissors, Feldmann and his team snipped out the gene for VSV's own outer surface protein—the key that made the virus target livestock—and replaced it with the single gene for the outer surface coat of Zaire Ebola virus (rVSV-ZEBOV):

- **A harmless chassis in an enemy coat**: Inside the particle, every internal gear and engine part belonged to the harmless livestock virus, so it could never cause Ebola. On the outside, it wore a coat of real Ebola surface proteins, tricking human immune scouts into thinking Ebola had arrived.
- **A fast seven-day shield**: Because the livestock virus chassis multiplied briskly for a few days inside muscle cells at the injection site and rang a strong innate alarm, a single shot raised protective Ebola antibodies within seven to ten days—fast enough to deploy in the middle of an active emergency.
- **Ring vaccination (`Ebola Ça Suffit!`)**: In March 2015 in coastal Guinea, World Health Organization teams led by Ana Maria Henao-Restrepo, Ira Longini, and smallpox veteran Donald Henderson revived the smallpox ring strategy. Whenever a new Ebola patient was diagnosed, field teams traced everyone who had touched the patient or their clothing, plus every contact of those contacts—an average ring of eighty people—and randomized 90 rings totaling 7,651 people to receive the shot either immediately or after a 21-day delay.

The results published in *The Lancet* in July 2015 were dramatic. In the delayed rings, sixteen people caught Ebola while waiting for their shot. Among the 2,119 contacts in the immediately vaccinated rings, once ten days had passed for immunity to kick in, zero cases of Ebola occurred—a measured ring efficacy of 100 percent. When a new Ebola outbreak struck the eastern Democratic Republic of the Congo between 2018 and 2020, ring vaccination of more than 300,000 contacts and frontline health workers with the Winnipeg Trojan horse (`Ervebo`) cut the death risk among exposed contacts by nearly 98 percent and turned a once-untreatable plague into a containable emergency.

CHAPTER 12

# Prefusion Spike and the Fat Bubble

*How Barney Graham, Jason McLellan, and Pieter Cullis Pinned a Shape-Shifting Protein Inside a Lipid Nanoparticle*

In 1966 in Washington, D.C., researchers tested a formalin-killed vaccine against Respiratory Syncytial Virus (RSV)—the leading cause of infant pneumonia—expecting it to work just like Jonas Salk's killed polio shot. Instead, it caused a medical disaster. When vaccinated toddlers caught wild RSV that winter, 80 percent had to be hospitalized with severe lung inflammation, and two children died. For nearly fifty years, nobody could explain why showing the immune system a dead RSV virus made the disease worse instead of better.

Between 2012 and 2016 at the National Institutes of Health Vaccine Research Center in Bethesda, Maryland, viral immunologist Barney Graham and structural biologist Jason McLellan solved that fifty-year mystery using X-ray crystallography. Viruses like RSV and coronaviruses use a spring-loaded surface protein—called the fusion (F) protein on RSV or the Spike protein on coronaviruses—to break into human cells. Before the virus touches a cell, that protein sits folded up like a cocked mousetrap in its "prefusion" shape. The moment it touches a target cell—or gets bumped by formaldehyde or heat in a vaccine vial—the protein snaps into a long, rigid spear in its "postfusion" shape to harpoon the cell membrane.

Graham and McLellan discovered that the 1966 formalin bath had tripped all the viral mousetraps in the vial before injection. Vaccinated children made millions of antibodies against the spent postfusion spear, which could not stop live viruses whose proteins

were still in the cocked prefusion shape. By mapping the atomic structure of coronavirus spikes in 2016 and 2017, McLellan and Graham found a simple mechanical fix. Swapping just two amino acids at the hinge of the spike for rigid proline rings (2P) acted like molecular cotter pins. Those two pins bolted the mousetrap in its live prefusion shape so the immune system made fifty times more neutralizing antibodies.

Delivering that pinned-spike blueprint into human cells as messenger RNA (mRNA) required three inventions working together:

- **The two-proline hinge pin (2P)**: Designed by McLellan and Graham in 2017 after studying the MERS coronavirus, two proline substitutions lock the spike protein in its live prefusion shape after your muscle cells read the mRNA recipe.
- **The stealth RNA letter**: Discovered by Katalin Karikó and Drew Weissman at the University of Pennsylvania in 2005, replacing the standard RNA letter uridine with a natural twin called pseudouridine (N1-methylpseudouridine) stops immune sensors from destroying the injected mRNA before your cells can read it.
- **The pH-switching fat bubble**: Engineered over forty years by Pieter Cullis and colleagues in Vancouver, a microscopic droplet made of four fats (a lipid nanoparticle) uses an ionizable fat that is positively charged in acidic mixing vats so it wraps tightly around negatively charged mRNA, turns neutral at pH 7.4 in your blood so it is non-toxic, and flips positive again inside the acidic bubble of a cell to pop the bubble open and release the mRNA.

When Chinese scientists posted the genetic sequence of SARS-CoV-2 online on January 10, 2020, Graham and McLellan slotted the 2P proline pins into the new spike sequence that weekend. Packaged inside modified mRNA and lipid nanoparticles, the first

clinical vials arrived at the NIH in forty-two days and achieved 94 to 95 percent efficacy against symptomatic illness across more than 73,000 trial volunteers later that year—while the exact same prefusion-pin discovery finally produced the world's first safe RSV vaccines in 2023.